

TEZPUR UNIVERSITY
Mid-Semester Examination, Autumn 2023
CO401: Artificial Intelligence

Full Marks: 40

Time: 2 hrs

Figures on the right indicate full marks for the individual questions.

1. Write short answers for the following (ANY FOUR): [3*4=12]
 - a. Differentiate between competitive multi-agent environment and cooperative multi-agent environment. Give examples of each.
 - b. Explain why the hill-climbing algorithm suffers from the *local maxima* problem.
 - c. Explain how the *stochastic local beam search* differs from *local-beam search* algorithm.
 - d. Explain how *alpha-beta pruning* improves the Minimax algorithm.
 - e. Explain the use of a *fitness function* in genetic algorithm.

2. For the 8-puzzle problem consider the following heuristics:

$h_1(n)$ = number of misplaced tiles in state n when compared to the goal state,

$h_2(n)$ = sum of the Manhattan distances of the misplaced tiles in state n from their correct positions in the goal state.

 - a. Using an example, show that $h_1(n)$ is a consistent heuristic. [4]
 - b. Explain why the heuristic $h_2(n)$ is better than $h_1(n)$. [3]

3. Given that a Propositional Logic knowledge base has the following sentences:

$T,$	$T \rightarrow C,$	$(C \wedge M) \rightarrow B,$
$I \rightarrow C,$	$(C \wedge F) \rightarrow G,$	F

Apply resolution refutation to prove that the knowledge base entails G . [8]

4. Define your own vocabulary and write the following as a well-formed First Order Logic formula. [3]

Every student must enroll in at least one course. A student must pass in all the enrolled courses to graduate.

5. Consider the mini-Sudoku problem in Figure (i).

The goal is to fill the blank spaces with digits 1 to 4 such that all four digits appear in each of the rows, columns, and inner boxes (shown using thick outlines) *exactly once*. Assuming each blank space corresponds to a variable, and the domain of each variable is the set $\{1,2,3,4\}$, answer the following:

- a. Formulate the constraints in mathematical form and draw the corresponding *constraint graph*.

[3]

- b. Apply forward checking algorithm to find a solution to the given mini-Sudoku problem. [7]

1	2	3	4
3	4	2	1
2	1	4	3
4	3	1	2

Figure (i)